Locating Failure in Visually Rich Document Understanding Systems: an Empirical Attribution

Lewei Xu[1] **Yihao Ding**[1*] **Zihan Xu**[2]
Daniel Yitian Su[1] **Siwen Luo**[1] **Daochang Liu**[1] **Yifan Peng**[3] **Wei Liu**[1]

[1]The University of Western Australia, Perth, Australia
[2]The University of Melbourne, Melbourne, Australia
[3]Weill Cornell Medicine, New York, USA
[*]*Correspondence:* yihao.ding@uwa.edu.au

Abstract

Multi-page visually rich document understanding (MP-VRDU) systems can fail when evidence is not faithfully represented, not selected, or not used correctly, yet final accuracy alone does not reveal where failures occur. We present an empirical attribution framework that examines these sources through controlled interventions in a single-pass retriever–generator pipeline. Experiments on MMLongBench-Doc, with cross-dataset and cross-family checks, show that complementary modalities can reduce sensitivity to conversion errors: images mitigate parser errors, while extracted text reduces dependence on visual resolution. Adding related non-gold pages can both repair incorrect answers and break correct ones, even when all annotated evidence is present, revealing instability beneath aggregate accuracy. Abstention trades unsupported answers for false refusals, while its interaction with reasoning instructions differs across backbones. Qualitative examples and accuracy–cost comparisons connect these findings to practical guidance on representation, evidence selection, and reasoner choice under deployment constraints.

1 Introduction

Multi-page visually rich documents combine text, tables, figures, and layout across many pages. They are common in finance, healthcare, and scientific research, where processing is often subject to privacy and compute constraints (Ma et al., 2024; Zhu et al., 2026). Multi-page visually rich document understanding (MP-VRDU) remains challenging because the relevant evidence may span text and figures, be difficult to locate across pages, or scattered across pages that must be combined (Ma et al., 2024; Deng et al., 2025). These challenges correspond to three core stages, *representation*, *selection*, and *reasoning*. (Cho et al., 2024; Han et al., 2025; Jain et al., 2025). Failures at different stages, however, can produce the same wrong answer, so final accuracy alone cannot reveal whether an error stems from lossy representation, inadequate selection, or flawed reasoning.

Existing studies partly address this by showing that MP-VRDU performance varies with input modality, parser, retrieval setting, and supplied evidence (Ma et al., 2024; Deng et al., 2025; Ak et al., 2026; Feng et al., 2026). However, these comparisons often show *where* performance drops rather than *why*: several factors often change together, and the same stage can fail for different reasons, such as missing required evidence or distracting pages. Mechanism-level studies examine Optical Character Recognition (OCR) errors and distractors, but mainly in text-based Retrieval-Augmented Generation (RAG) and one mechanism at a time (Zhang et al., 2025; Cuconasu et al., 2024). Without clearer attribution, it is unclear how a given MP-VRDU system should be improved. Key MP-VRDU design choices therefore still lack a mechanism-level empirical basis, particularly for privacy-sensitive and resource-constrained local deployment.

We aim to explain why MP-VRDU systems fail, in terms that can directly inform system design. We trace errors through three loci of the evidence flow (representation, selection, and reasoning) and use controlled interventions to separate failures that would otherwise appear as the same wrong answer. The main analysis uses MMLongBench-Doc (Ma et al., 2024) with Qwen3-VL (Team, 2025). LongDocURL (Deng et al., 2025) and Gemma-3 (Team et al., 2025) provide complementary checks across document collections and model families. We also consider memory, latency, and compute, which constrain systems that must process privacy-sensitive documents locally.

Our analysis reveals several design trade-offs. For representation, perfecting a single modality is often unnecessary. Complementary inputs substantially reduce sensitivity to parser and resolution errors. Adding page images shrinks the accuracy

spread across parsers from 15.7 to 4.2 points on born-digital documents, and embedded text with page images matches parser text with page images while using 28% fewer input tokens. For selection, retrieval rank does not reliably indicate which required evidence can be safely omitted, and added context can reduce accuracy even when coverage is already complete, particularly for multi-hop questions. For reasoning, scaling improves overall accuracy but does not narrow the gap between single- and multi-hop questions. Abstention instructions reduce unsupported answering at the cost of false refusals; reasoning before abstaining largely recovers answerable accuracy for Qwen3-VL but offers little benefit for Gemma-3. Richer inputs, deeper retrieval, and larger models are therefore not universal fixes. The results instead motivate selective representation, evidence-set control, targeted reasoning, and backbone-specific calibration under deployment constraints.

Our contributions are as follows: (1) We organize MP-VRDU failures into three architecture-independent loci: representation, selection, and reasoning (§3). **(2)** We investigate these failure modes through controlled interventions and stratified comparisons in a unified pipeline, with experiments across document collections and independently developed model families (§4–§5). **(3)** We connect the findings to qualitative examples and accuracy–cost comparisons to derive practical guidance for representation, retrieval, and reasoner choices under task requirements and resource constraints (§6).

2 Related Work

Dataset and Diagnostic Studies. MMLongBench-Doc compares page-image and OCR-text inputs, reports performance by document domain, evidence type and position, and single- versus cross-page questions, and supplements oracle-page experiments with qualitative error categorisation (Ma et al., 2024). LongDocURL evaluates understanding, numerical reasoning, and cross-element locating across document sources and evidence configurations, compares image-input strategies and text parsers, and analyses errors involving perception, reasoning, incomplete evidence, and output formatting (Deng et al., 2025). DocScope separately evaluates page localisation, region grounding, fact extraction, and answer verification, using oracle interventions to identify intermediate bottlenecks (Feng et al., 2026). Complementing these benchmark analyses, Ak et al. (2026) compare input modalities, retrieval strategies, and agentic access, examining accuracy–cost trade-offs across model capacities and benchmarks. The mechanism-level basis for choosing representation, selection, and reasoning configurations remains incomplete; we examine these choices through controlled interventions, qualitative examples, and deployment-cost comparisons.

Failure Mechanisms in RAG. Mechanism-specific studies examine conversion errors, evidence composition, and response calibration. OHRBench perturbs semantic and formatting aspects of document parses to measure their effects on retrieval and generation (Zhang et al., 2025). The Power of Noise examines passage relevance, position, and number, finding that highly ranked irrelevant passages can harm accuracy while random additions can help (Cuconasu et al., 2024). Jin et al. (2025) attribute degradation at greater retrieval depths to hard negatives and evaluate retrieval reordering and fine-tuning as mitigations. Context-faithful Prompting improves contextual adherence and abstention through opinion-based prompts and counterfactual demonstrations (Zhou et al., 2023). Sufficient Context separates insufficient evidence from failures to use sufficient evidence, and combines sufficiency with model confidence for selective generation (Joren et al., 2025). Although OCR effects have been studied on document benchmarks, these selection and abstention analyses primarily evaluate text-based tasks; we examine representation, selection, and reasoning mechanisms together through controlled interventions in multi-page visually rich document understanding.

3 Attribution Framework

We decompose MP-VRDU design into three fundamental loci, defined independently of system architecture: **representation**, **selection**, and **reasoning**. Every MP-VRDU technique targets failure at one or more of them. The decomposition thus connects architectural choices to the failures they aim to reduce. Figure 1 summarises the loci and their mechanisms.

3.1 Representation

Representation determines what information from the document is made available to later stages,

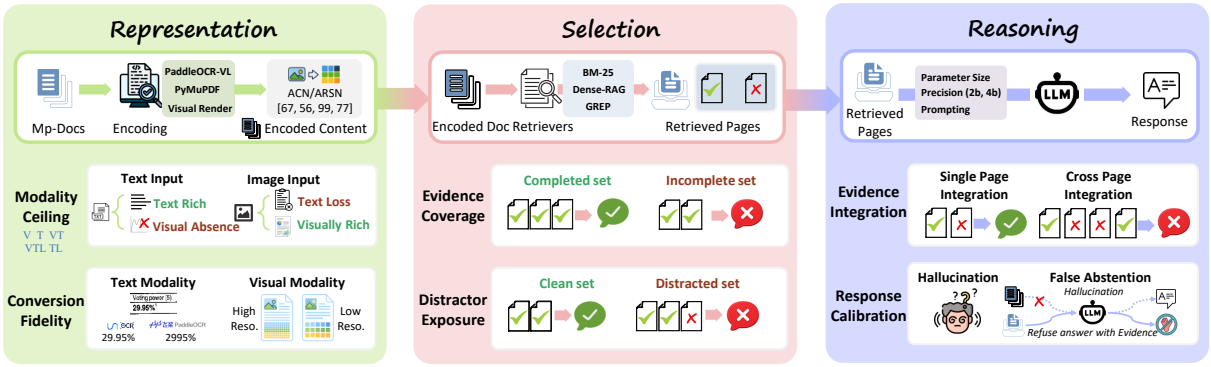


Figure 1: The three failure loci and six mechanisms.

such as embedded text, parser markdown, rendered pages, or visual crops. Changing the form of the same content is a representation choice; choosing which content to include belongs to selection, and using that content belongs to reasoning. Within this locus, failures arise in two ways. **(1) Modality Ceiling.** A representation limits which signals are accessible at its granularity: text may omit non-linguistic visual information, and images require sufficient resolution to expose fine detail. **(2) Conversion Fidelity.** Conversion may degrade information that the modality could otherwise preserve, for example through OCR errors or corrupted table structure (Zhang et al., 2025).

3.2 Selection

Selection determines which represented content is placed before the reasoner within the available context budget. It may operate over pages, sections, tables, or visual crops, in a single pass or through repeated acquisition. **(1) Evidence Coverage.** Coverage is the proportion of required evidence units included in the selected context; omission leaves the reasoner without information the answer needs. **(2) Distractor Exposure.** Exposure is the irrelevant or misleading content that accompanies the required evidence. It makes the relevant information harder to identify and use, even when coverage is complete.

3.3 Reasoning

Reasoning maps the supplied evidence and instruction to a response: a final answer or structured output. When representation preserves the required evidence and selection supplies it, the remaining failures concern how the reasoner interprets and acts on that evidence. Although such failures take many forms, we focus on two that are particularly relevant to MP-VRDU. **(1) Evidence Integration.** The reasoner may process individual evidence units correctly yet fail to combine them across pages or sources; long-context position effects can amplify this difficulty (Liu et al., 2024). **(2) Response Calibration.** The decision to answer should reflect evidential support. Failure includes both unsupported answering (hallucination) and false abstention despite sufficient evidence. The two must be assessed jointly, because greater willingness to answer can reduce false abstention while increasing unsupported responses (Whitehead et al., 2022).

4 Methodology

We test the six failure mechanisms through controlled experiments and derive design implications from the results. Appendix A provides full experimental specifications.

4.1 Attribution by Construction

We instantiate the framework in a **single-pass retriever-generator** document question-answering pipeline whose stages align with the three loci. *Encoding* determines the representation available to the system, *retrieval* determines which pages reach the reasoner, and *inference* determines how the supplied evidence is used to answer. Encoding converts each page to parser markdown, a rendered image, or both. Retrieval scores pages against the question and selects the top k at page granularity; oracle conditions instead supply the annotated gold pages. Inference gives a VLM only the selected pages, the question, and a task instruction, and produces an answer in one pass. Because the stages run once and in order, each comparison can change the representation, evidence set, or reasoning condition relevant to one mechanism while holding the other stages fixed.

4.2 Experiments

Representation. To test the *modality ceiling*, we supply gold pages as embedded text extracted by PyMuPDF (**T**), PaddleOCR-VL markdown (**TL**) (Cui et al., 2025), markdown plus page images (**TLV**), or images alone (**V**), and compare results by evidence source. To test *conversion fidelity*, we compare PaddleOCR-VL with MinerU 2.5 (Niu et al., 2026) and Unlimited-OCR (Yin et al., 2026) within **TL** and **TLV**. Separately, we sweep the `low`, `med`, and `high` image presets (Appendix A.2) within **V** and **TLV**. The parser and resolution changes are evaluated separately. The default is **TLV** with PaddleOCR-VL at `med`: pages are rendered at 200 DPI and downsampled to at most 501,760 pixels per page (a median of 456 visual tokens).

Selection. To test *evidence coverage*, we begin with the complete gold-page set of each multi-page question. ColQwen3, a ColPali-style visual retriever (Faysse et al., 2025), ranks every page against the question. We remove either the highest- or the lowest-ranked gold page to test whether retrieval rank predicts the cost of omission, and then retain only one of those pages to examine more severe coverage losses. To test *distractor exposure*, we retain the complete gold set and add the top k non-gold pages by retrieval rank, for $k = 0, \ldots, 18$. This constructs an idealised high-recall retrieval setting: annotated evidence coverage remains complete while the amount of additional retrieved context increases. The experiment measures the overall effect of this additional context, including its content, length, and changes in evidence position, rather than isolating these factors individually.

Reasoning. The default reasoner is Qwen3-VL-8B-Instruct (Team, 2025) in `bf16` with greedy decoding. To examine *evidence integration*, we supply gold pages and compare accuracy by evidence-page and evidence-source count across Qwen3-VL sizes from 2B to 32B. To test *response calibration* with the 8B model, we compare four prompt modes (`none`, `cot`, `abstain`, and `abstain_cot`) on answerable questions with gold pages and unanswerable questions with BM25's top three pages. `cot` requests step-by-step reasoning followed by a final answer prefixed with "Answer:". `abstain` requests the exact response "Not answerable" when the question cannot be answered from the evidence provided. `abstain_cot` combines both instructions. The output limit is 256 tokens, or 2048 for prompts that request CoT.

Generalization. To check whether the findings extend beyond one model family, we use Gemma-3 (Team et al., 2025), which was developed independently of Qwen3-VL, with different training data and development pipelines. On MMLongBench-Doc, we repeat the distractor-exposure and response-calibration experiments with Gemma-3-12B, and the evidence-integration experiment across Gemma-3 sizes.

4.3 Dataset and Evaluation

MMLongBench-Doc (Ma et al., 2024) contains 135 multimodal documents and questions annotated with evidence pages, evidence sources, and answerability. It is the only benchmark we are aware of whose annotations support all of the representation, selection, and reasoning experiments. LongDocURL (Deng et al., 2025) provides a complementary test of the modality-ceiling findings on exclusively born-digital documents, where answer-bearing content is often directly accessible in the text layer, even for figure questions. Its questions are generated by LVLMs and verified by humans, which may favor evidence recoverable through that construction pipeline, and its annotated evidence pages need not form a minimal sufficient set. We therefore do not repeat the conversion-fidelity and evidence-coverage analyses on LongDocURL.

We use Gemini-2.5-Flash (Comanici et al., 2025) at temperature zero as the answer judge. Given only the question, the gold answer, the unanswerable flag, and the extracted model answer, it returns `correct`, `incorrect`, or `abstained`. Any refusal is labeled `abstained` regardless of answerability, and a refusal is credited as correct only for an unanswerable question. Accuracy is the proportion of responses credited as correct, and paired transitions track correct-to-incorrect and incorrect-to-correct rates. On validation subsets, the judge agrees with GPT-4o on 96.0% of verdicts and with a human annotator on 98.3%.

5 Empirical Findings

This section reports the controlled results for each locus under the setup of §4.

5.1 Representation

Modality Ceiling: Evidence Accessibility Depends on Modality. A modality can only sup-

Evidence	Abstention				Accuracy			
	T	TL	TLV	V	T	TL	TLV	V
MMLongBench-Doc								
Text	23.0	11.5	7.3	8.0	39.2	47.8	**55.3**	44.3
Layout	20.3	10.2	0.8	4.2	28.0	37.3	**50.8**	44.1
Table	15.9	7.2	6.7	**9.1**	40.6	**48.4**	**48.4**	37.8
Chart	32.6	**28.1**	7.3	6.2	18.5	19.1	**45.5**	43.3
Figure	**36.6**	24.8	**7.6**	7.9	15.5	23.8	**44.8**	40.3
All	26.3	17.2	6.2	7.1	31.9	38.8	**52.5**	45.6
LongDocURL								
Text	4.6	3.5	**1.5**	**1.6**	75.3	79.5	**82.3**	74.0
Layout	2.1	1.6	0.7	0.9	48.7	52.4	**56.3**	52.2
Table	5.0	3.2	1.1	0.8	61.2	**69.5**	68.2	57.3
Figure	**6.7**	**8.6**	**1.5**	1.3	62.5	65.7	**73.4**	65.2
All	4.7	4.3	1.3	1.3	63.7	68.0	**71.9**	63.1

Table 1: Abstention and accuracy by evidence source and representation.

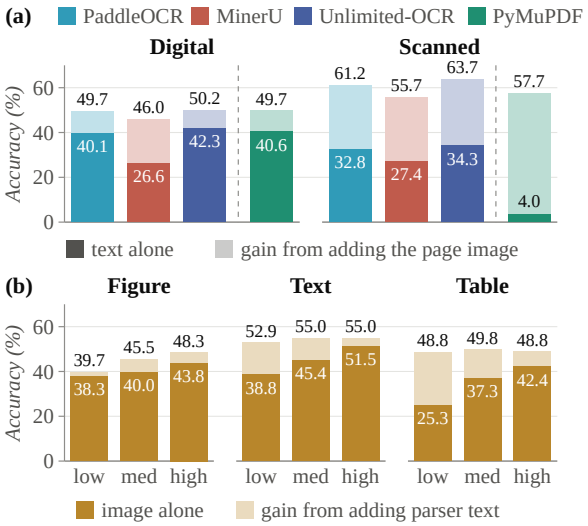


Figure 2: **Representation: Conversion Fidelity** (a) Accuracy by text source and scan status. (b) Accuracy by image resolution, with and without parser text.

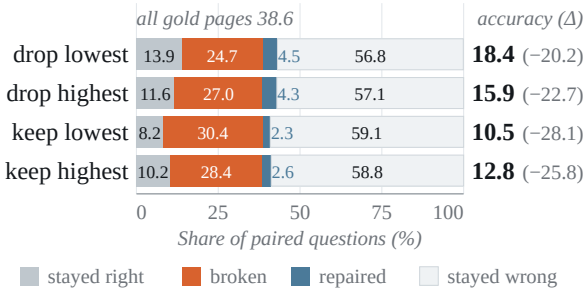


Figure 3: Paired verdict transitions and accuracy after removing gold pages on MMLongBench-Doc.

port questions whose required evidence it preserves. On MMLongBench-Doc, this effect is clearest for chart and figure questions (Table 1): **T** and **TL** refuse them far more often, whereas adding vision brings abstention below 10% across evidence sources. The converse also holds: **V** remains less accurate than **TLV** when answers depend on dense text or precise numerical values. Text and vision therefore preserve complementary evidence. LongDocURL shows the same dependency in a different document setting. Its born-digital only documents preserve more evidence in text, so adding vision to **TL** improves overall accuracy by only 3.9 points, while figure questions gain 7.7 points. The modality ceiling therefore depends on whether the evidence required by a question is accessible in the chosen modality.

Conversion Fidelity: Complementary Modalities Reduce Conversion Sensitivity. Conversion errors matter less when the affected information remains accessible through another modality. Page images can recover information lost during parsing, while parser text can reduce the need for high-resolution images to read fine-grained text (Figure 2). Adding page images reduces the accuracy spread across parsers from 15.7 to 4.2 points on born-digital documents. Increasing image resolution brings substantially larger gains under **V** than under **TLV** on text and table questions (17.1 vs. 0.0 points on tables from `low` to `high`). The compensation is incomplete where the modalities preserve non-substitutable information. Scanned documents remain sensitive to parser quality, and figure questions to visual resolution. Conversion fidelity therefore matters most when degraded information cannot be recovered from a complementary modality.

5.2 Selection

Evidence Coverage: Omission Causes Large Accuracy Losses. Accuracy is highly sensitive to missing required evidence. Removing a single gold page reduces accuracy from 38.6% to 18.4% when the lowest-ranked page is removed, and to 15.9% when the highest-ranked page is removed (Figure 3). Paired verdicts show that correct answers are broken far more often than incorrect answers are repaired, so the remaining evidence rarely compensates for the omission. The similar losses under both removals also show that retrieval rank does not reliably predict the cost of excluding a required page. On MMLongBench-Doc, coverage failure therefore depends primarily on whether the necessary evidence is present, not on where it appears in the ranking.

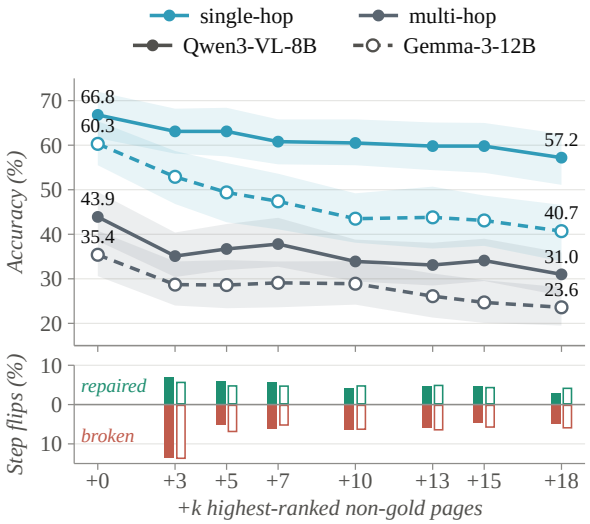


Figure 4: Accuracy and paired verdict transitions as non-gold pages are added to complete gold-page sets.

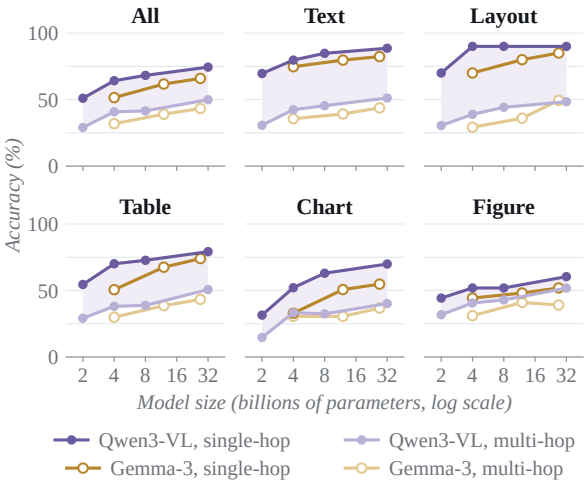


Figure 5: Accuracy on single- and multi-hop questions across Qwen3-VL model scales.

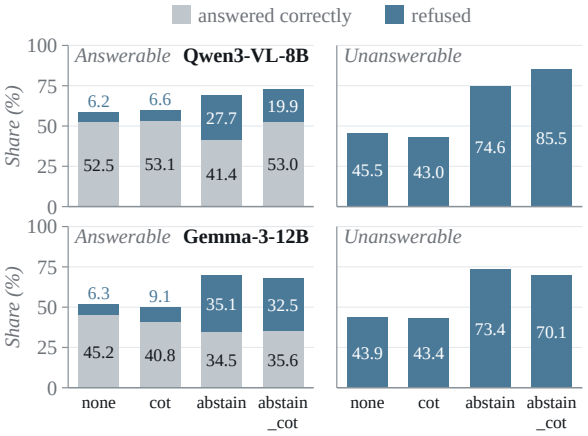


Figure 6: Answerable accuracy, anwerable refusal and unanswerable refusal across four prompt modes.

Distractor Exposure: Multi-Hop Accuracy Drops Sharply with Added Context. Adding non-gold pages generally reduces accuracy, and multi-hop questions show the sharpest early degradation (Figure 4). The pattern holds for both backbones. For Qwen3-VL-8B, multi-hop accuracy drops from 43.9% to 35.1% after only three added pages, whereas single-hop accuracy declines gradually from 66.8% to 57.2% across the sweep. Gemma3-12B shows the same overall trend, with multi-hop accuracy falling from 35.4% to 23.6%, but its single-hop accuracy is more sensitive to added context, decreasing from 60.3% to 40.7%. Because the complete gold set remains available throughout, these losses reflect difficulty using the required evidence amid competing context, not missing evidence. Paired transitions show that added pages act in both directions: they fix some incorrect answers by adding useful information, but break some correct answers by introducing competing information. Distractor exposure therefore makes it harder to use the right evidence consistently, and the sensitivity differs across backbones.

5.3 Reasoning

Evidence Integration: Scaling Helps, but the Multi-Hop Gap Persists. Larger reasoners improve both single- and multi-hop accuracy, but do not close the gap between them (Figure 5). For Qwen3-VL, multi-hop accuracy rises from 29.0% at 2B to 49.9% at 32B, and single-hop accuracy from 51.0% to 74.4%. So the gap remains roughly constant (22.0 points at 2B, 24.5 at 32B). Scaling thus improves evidence use in general rather than the specific difficulty of combining evidence across pages or sources. Gemma-3 shows the same separation between single- and multi-hop accuracy, although its gains vary across evidence types. One likely explanation is that scaling improves processing of individual evidence units and multi-hop reasoning, while multi-hop questions still require additional operations to locate, align, and combine information across sources. Thus, greater capacity raises overall performance, but does not remove the extra burden of evidence integration.

Response Calibration: Reasoning Before Abstaining Limits False Refusal. Abstention prompting substantially increases refusal on unanswerable questions, but it also makes the reasoner more likely to refuse answerable ones (Figure 6). This trade-off appears for both backbones. For Qwen3-VL-8B, `abstain` raises unanswerable re-



Figure 7: Qualitative example of representation, selection, and reasoning failures and the design response to each.

fusal from 45.5% to 74.6%, but also increases false refusal from 6.2% to 27.7% and reduces answerable accuracy from 52.5% to 41.4%. Gemma3-12B shifts the same way, with unanswerable refusal increasing from 43.9% to 73.4% and false refusal from 6.3% to 35.1%. Adding CoT changes this balance differently for the two models. For Qwen3-VL-8B, `abstain_cot` reduces false refusal to 19.9%, raises answerable accuracy to 53.0%, and improves unanswerable refusal further to 85.5%. For Gemma3-12B, false refusal falls to 32.5%, but answerable accuracy improves by only 1.1 points and unanswerable refusal declines to 70.1%. Reasoning before abstaining can therefore recover supported answers, but the benefits depends on the backbone, and false refusal remains substantially above on-instruction baseline for both models.

6 Practical Discussion

We now discuss what the findings of §5 imply for designing and deploying document question-answering systems.

6.1 Representation

Representation failures arise when the input does not preserve task-relevant information andcomplementary modalities can recover missing evidence and reduce sensitivity to conversion errors. Figure 7a illustrates two such failures: text-only input loses visual attributes such as icon colour, and parser errors distort textual evidence that remains recoverable from the page image. This compensation works in both directions: images can mitigate parsing errors, while extracted text can reduce dependence on visual resolution when it preserves the required information. However, adding more modalities does not always help. Embedded text with images (TV) achieves accuracy comparable to parser text with images (TLV), 51.6% versus 52.5%, with fewer tokens per page (959 versus 1,345; Table 2a). Visual processing also adds substantial prefill latency (26.3s for TLV versus 1.8s for TL).

Deployment suggestions. Systems should reply on embedded text where where it is reliable, add page images when visual information is needed or extraction is unreliable, and re-parse when the existing text is inadequate. For local deployment, modality choice, conversion fidelity, and evidence selection depth should be jointly configured to balance accuracy, memory, and inference latency.

6.2 Selection

Selection failures arise when necessary evidence is omitted or when competing context interferes with its use. Selection should therefore aim for

(a) Representation	Acc.	Text	Vis.	Tokens	Prefill	Decode
T	31.9	501	0	501	1.1	5.4
TL	38.8	886	0	886	1.5	6.3
TV	51.6	500	459	959	23.8	6.5
TLV	52.5	886	459	1345	24.2	6.6
V	45.6	20	459	479	22.7	6.6
(b) Retriever	**R@1**	**R@5**	**R@10**	**Idx. (s)**	**Qry. (ms)**	**GB**
BM25	32.7	58.1	68.1	0.02	0.3	CPU
BGE-M3	32.8	61.3	72.4	0.01	47	1.2
Qwen3-Emb.	33.5	63.9	74.8	0.19	89	13.1
ColModernVBERT	47.1	74.0	82.7	0.91	41	5.1
ColQwen2.5	49.2	77.2	84.8	1.50	293	8.6
ColQwen3	53.7	81.0	87.9	0.57	110	9.4
(c) Reasoner	**T**	**TL**	**TLV**	**V**	**GB**	**Time (s)**
Qwen3-VL-8B, bf16	31.9	38.4	52.4	45.5	17.5	30.8
8-bit	32.5	37.9	53.6	45.7	10.3$^{\sim}$	43.5
4-bit	31.8	39.4	51.8	45.5	6.8$^{\sim}$	35.4
Qwen3-VL-32B, 4-bit	37.0	42.1	59.1	52.3	20.0$^{\sim}$	47.5†
InternVL3-8B	19.3	25.3	32.6	24.1	15.9	5.3
MiniCPM-V-4.5	30.7	38.1	60.1	53.3	17.4	47.2
GLM-4.6V-Flash	25.9	32.9	42.7	30.9	20.6	19.8
Llama-3.2-11B-V	26.5	34.3	33.3	25.3	21.3	10.4
Gemma-3-12B	30.5	36.9	48.1	41.6	24.4	8.0

Table 2: Accuracy and cost of each representation, retriever and reasoner.

collective evidence coverage while controlling distractor exposure. Figure 7b illustrates both failures: without a complementary page the reasoner cannot integrate the evidence, and added context can either disrupt or restore an answer. These changes occur even when all annotated gold pages remain present, and successive repair and break transitions reveal instability beneath aggregate accuracy. The findings agree with prior observations on distracting passages and hard negatives in RAG (Cuconasu et al., 2024; Jin et al., 2025), but they also show that related non-gold pages can help provide useful context to interpret evidence. A plausible answer from a retrieved subset therefore does not establish that further document context is unnecessary. Structured navigation, summary-based reranking, and iterative retrieval are possible ways to balance these competing requirements (Sun et al., 2025; Jain et al., 2025). Table 2b further shows that ColQwen3 improves R@5 over BM25 (81.0% versus 58.1%), but requires GPU memory and has higher query latency (110 versus 0.3 ms).

Deployment suggestions. For frequently queried document collections, stronger visual retrievers may justify their indexing and memory costs, whereas lightweight text retrievers remain attractive for resource-constrained or infrequently queried collections. Since retrieval and reasoning costs recur with every query, retrieval depth and refinement strategies should be evaluated by end-to-end answer quality, stability, and latency rather than recall alone.

6.3 Reasoning

The reasoning findings reveal two distinct failures: models struggle to integrate evidence across multiple sources, and abstention instructions can suppress answers supported by the available evidence (Figure 7c). Abstention reduces unsupported answering at the cost of false refusals; adding reasoning recovers more supported answers for Qwen3-VL than for Gemma-3. These failures call for targeted strategies for evidence integration and backbone-specific response calibration. Self-Consistency (Wang et al., 2023) and evidence-based verification may improve multi-step reasoning and help distinguish insufficient evidence from reasoning errors, although their effectiveness in this setting remains to be evaluated.

Deployment suggestions. Reasoning and abstention instructions should be tuned jointly on the intended task, reporting answerable accuracy, unsupported answering, and false refusals separately. Accuracy–cost trade-offs differ substantially across model families and precision settings (Table 2c). Backbone selection should consider task-specific accuracy against inference speed. Quantisation reduces GPU memory but can increase latency. Additional reasoning and verification can be reserved for complex or uncertain cases to manage computational costs.

7 Conclusion

Our controlled quantitative comparisons and qualitative examples connect MP-VRDU failures to practical design choices. Images can compensate for parsing errors, while extracted text can reduce dependence on visual resolution when it preserves the required information. Related non-gold pages can repair or break answers despite complete annotated coverage, revealing context-dependent instability beneath aggregate accuracy. Reasoning and abstention instructions interact differently across backbones, requiring supported answering and false refusal to be evaluated together. These findings inform representation, evidence selection, and reasoner choices under resource constraints. Future work should validate adaptive policies that acquire complementary representations, expand or refine evidence sets, and invoke further reasoning or verification, measuring both supported-answer quality and the total cost of those decisions.

Limitations

Our evaluation focuses on MMLongBench-Doc, with additional experiments on LongDocURL. The findings may therefore be influenced by benchmark characteristics, including document types, question distributions, and evidence annotations. Although we evaluate multiple models, the main failure-mechanism analyses focus on Qwen3-VL and Gemma-3, and the observed patterns may differ across other model families. Our experiments also use a single-pass retriever–generator pipeline, leaving interactions in iterative and agentic systems unexplored. Finally, our evidence selection is examined at page granularity rather than at finer levels such as passages or visual regions.

References

Kenan E. Ak, Jay Mohta, Gwang Gook Lee, Yan Xu, and Dimitrios Dimitriadis. 2026. An empirical study of VLM pipelines for long-document QA. *CoRR*, abs/2609.29933.

Jaemin Cho, Debanjan Mahata, Ozan Irsoy, Yujie He, and Mohit Bansal. 2024. M3docrag: Multi-modal retrieval is what you need for multi-page multi-document understanding. *CoRR*, abs/2411.04952.

Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, and Evan Rosen. 2025. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *Preprint*, arXiv:2507.06261.

Florin Cuconasu, Giovanni Trappolini, Federico Siciliano, Simone Filice, Cesare Campagnano, Yoelle Maarek, Nicola Tonellotto, and Fabrizio Silvestri. 2024. The power of noise: Redefining retrieval for RAG systems. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, pages 719–729. ACM.

Cheng Cui, Ting Sun, Suyin Liang, Tingquan Gao, Zelun Zhang, Jiaxuan Liu, Xueqing Wang, Changda Zhou, Hongen Liu, Manhui Lin, Yue Zhang, Yubo Zhang, Handong Zheng, Jing Zhang, Jun Zhang, Yi Liu, Dianhai Yu, and Yanjun Ma. 2025. Paddleocr-vl: Boosting multilingual document parsing via a 0.9b ultra-compact vision-language model. *CoRR*, abs/2510.14528.

Chao Deng, Jiale Yuan, Pi Bu, Peijie Wang, Zhong-Zhi Li, Jian Xu, Xiao-Hui Li, Yuan Gao, Jun Song, Bo Zheng, and Cheng-Lin Liu. 2025. Longdocurl: a comprehensive multimodal long document benchmark integrating understanding, reasoning, and locating. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, pages 1135–1159. Association for Computational Linguistics.

Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. 2025. Colpali: Efficient document retrieval with vision language models. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.

Xiang Feng, Jiawei Zhou, Zhangfeng Huang, Kewei Wang, Shanshan Ye, Jinxin Hu, Zulong Chen, Yong Luo, and Jing Zhang. 2026. Docscope: Benchmarking verifiable reasoning for trustworthy long-document understanding. *CoRR*, abs/2605.08888.

Siwei Han, Peng Xia, Ruiyi Zhang, Tong Sun, Yun Li, Hongtu Zhu, and Huaxiu Yao. 2025. Mdocagent: A multi-modal multi-agent framework for document understanding. *CoRR*, abs/2503.13964.

Chelsi Jain, Yiran Wu, Yifan Zeng, Jiale Liu, Shengyu Dai, Zhenwen Shao, Qingyun Wu, and Huazheng Wang. 2025. Simpledoc: Multi-modal document understanding with dual-cue page retrieval and iterative refinement. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China, November 4-9, 2025*, pages 28410–28427. Association for Computational Linguistics.

Bowen Jin, Jinsung Yoon, Jiawei Han, and Sercan Ö. Arik. 2025. Long-context llms meet RAG: overcoming challenges for long inputs in RAG. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.

Hailey Joren, Jianyi Zhang, Chun-Sung Ferng, Da-Cheng Juan, Ankur Taly, and Cyrus Rashtchian. 2025. Sufficient context: A new lens on retrieval augmented generation systems. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Trans. Assoc. Comput. Linguistics*, 12:157–173.

Yubo Ma, Yuhang Zang, Liangyu Chen, Meiqi Chen, Yizhu Jiao, Xinze Li, Xinyuan Lu, Ziyu Liu, Yan Ma, Xiaoyi Dong, Pan Zhang, Liangming Pan, Yu-Gang Jiang, Jiaqi Wang, Yixin Cao, and Aixin Sun. 2024. MMLONGBENCH-DOC: benchmarking long-context document understanding with visualizations. In *Advances in Neural Information Processing*

Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024.

Junbo Niu, Zheng Liu, Zhuangcheng Gu, Bin Wang, Linke Ouyang, Zhiyuan Zhao, Tao Chu, Tianyao He, Fan Wu, Qintong Zhang, Zhenjiang Jin, Guang Liang, Rui Zhang, Wenzheng Zhang, Yuan Qu, Zhifei Ren, Yuefeng Sun, Zirui Tang, Boyu Niu, and 40 others. 2026. Mineru2.5: A decoupled vision-language model for efficient high-resolution document parsing. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track), ACL 2026, San Diego, California, USA, July 2-7, 2026*, pages 13–42. Association for Computational Linguistics.

Li Sun, Liu He, Shuyue Jia, Yangfan He, and Chenyu You. 2025. Docagent: An agentic framework for multi-modal long-context document understanding. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China, November 4-9, 2025*, pages 17701–17716. Association for Computational Linguistics.

Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, and 1 others. 2025. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*.

Qwen Team. 2025. Qwen3-vl technical report. *CoRR*, abs/2511.21631.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.

Spencer Whitehead, Suzanne Petryk, Vedaad Shakib, Joseph Gonzalez, Trevor Darrell, Anna Rohrbach, and Marcus Rohrbach. 2022. Reliable visual question answering: Abstain rather than answer incorrectly. In *Computer Vision - ECCV 2022 - 17th European Conference, Tel Aviv, Israel, October 23-27, 2022, Proceedings, Part XXXVI*, volume 13696 of *Lecture Notes in Computer Science*, pages 148–166. Springer.

Youyang Yin, Huanhuan Liu, Yy, Qunyi Xie, Chaorun Liu, Shiqi Yang, Shaohua Wang, Zhanlong Liu, Hao Zou, Jinyue Chen, Shu Wei, Jingjing Wu, Mingxin Huang, Zhen Wu, Guibin Wang, Tengyu Du, and Lei Jia. 2026. Unlimited OCR works. *CoRR*, abs/2606.23050.

Junyuan Zhang, Qintong Zhang, Bin Wang, Linke Ouyang, Zichen Wen, Ying Li, Ka-Ho Chow, Conghui He, and Wentao Zhang. 2025. OCR hinders RAG: evaluating the cascading impact of OCR on retrieval-augmented generation. In *IEEE/CVF International Conference on Computer Vision, ICCV 2025, Honolulu, HI, USA, October 19-25, 2025*, pages 17443–17453. IEEE.

Wenxuan Zhou, Sheng Zhang, Hoifung Poon, and Muhao Chen. 2023. Context-faithful prompting for large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023*, volume EMNLP 2023 of *Findings of ACL*, pages 14544–14556. Association for Computational Linguistics.

Dawei Zhu, Rui Meng, Jiefeng Chen, Sujian Li, Tomas Pfister, and Jinsung Yoon. 2026. Doclens: A tool-augmented multi-agent framework for long visual document understanding. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2026, San Diego, California, United States, July 2-7, 2026*, pages 26804–26829. Association for Computational Linguistics.

```
You are answering a question about a document.
{instruction}
Question:
{question}
Document evidence:
{context}
Answer:
```

Figure 8: Common single-pass reasoner template. The instruction comes from Table 5; the context contains the selected pages in document order and in the representation specified by the condition.

A Additional Experimental Details

This appendix supplements §4 with the prompts, settings and evaluation rubrics that the main text does not give.

A.1 Pipeline

The pipeline of §4.1 has three stages, each run once per question.

(1) Encoding. Each page is rasterised at 200 DPI, and its embedded text layer is extracted with PyMuPDF. A document parser converts the rendered page to markdown, one page at a time, and the result is stored so that every condition reads the same parse. Page images are downsampled to the pixel cap of the resolution preset before they reach the reasoner. The representation then determines which of these artifacts form the context (Table 3).

(2) Retrieval. A retriever scores every page of the question's document against the question and returns the top k pages. Text retrievers rank page text and visual retrievers rank page images. The controlled experiments replace this stage with constructed page sets: representation and reasoning conditions supply the annotated gold pages, and selection conditions remove gold pages or add non-gold ones. ColQwen3 is used only to order pages where a condition depends on rank, and BM25 only to supply three pages for unanswerable questions, which have no gold evidence.

(3) Inference. The reasoner receives the selected pages, the question and an instruction in the template of Figure 8, and answers in a single pass without tools or follow-up turns. Pages are restored to document order, whatever their rank. Under **TLV**, the text of all supplied pages forms one block, followed by their images in the same order. Decoding is greedy, with no sampling parameters and no cap on input tokens, so an output is determined by its question, page set, representation, reasoner, resolution and prompt mode.

Rep.	Text channel	Image channel
T	PyMuPDF embedded text (no parsing)	–
TL	PaddleOCR-VL markdown	–
TV	PyMuPDF embedded text (no parsing)	Page image (`med`)
TLV	PaddleOCR-VL markdown	Page image (`med`)
V	–	Page image (`med`)

Table 3: Page representations.

Channel	Setting	Detail
Text	PaddleOCR-VL (default)	0.9B
Text	MinerU 2.5	1.2B
Text	Unlimited-OCR	3.3B
Image	`low`	285 tok/page
Image	`med` (default)	456 tok/page
Image	`high`	744 tok/page

Table 4: Conversion-fidelity settings.

Comparisons. Every intervention is compared with its baseline on the same questions, and a question that cannot satisfy a condition is excluded from that comparison. A *single-hop* question has one gold page and cites one evidence source; every other answerable question is *multi-hop*.

A.2 Representation Settings

Table 3 defines the representations and Table 4 the fidelity settings. Parsers emit page-level markdown, and no layout coordinates reach the reasoner. **T** reads the PDF's embedded text layer, which is usually empty on scanned documents. The image presets cap a page at 313,600 (`low`), 501,760 (`med`) and 752,640 (`high`) pixels, and the token counts in Table 4 are Qwen3-VL's median visual tokens per page.

A.3 Prompt Modes

Table 5 gives the instruction for each prompt mode, including three that appear only in Appendix B. Both `cot` and `abstain` include the `grounded` sentence. Under the modes that request reasoning, the text after the last `Answer:` marker is extracted as the answer before judging.

Mode	Instruction
`none`	(empty)
`grounded`	Use only the provided document evidence and keep the answer concise.
`abstain`	`grounded`, plus: If the evidence does not contain the answer, answer exactly: Not answerable.
`abstain_balanced`	`abstain`, plus: If the evidence does contain the answer, give it: do not decline a question the evidence supports.
`cot`	`grounded`, plus: Think step by step before answering. Then write your final answer on a new line beginning with 'Answer:'.
`extract_cot`	`grounded`, plus a verbatim-evidence extraction step, with visual evidence described in enough detail to answer from the description alone, then reasoning constrained to what was extracted, then the `Answer:` line.
`abstain_cot`	`grounded`, plus: If the evidence does not contain the answer, answer exactly: Not answerable. Think step by step before answering. Then write your final answer on a new line beginning with 'Answer:'.

Table 5: Reasoning prompt modes.

A.4 Datasets

Tables 6 and 7 summarise the two benchmarks. On MMLongBench-Doc, a question is unanswerable when its gold answer is *Not answerable*, and a document is considered scanned when its pages carry no usable text layer. Evidence source is multi-label on both benchmarks, so a question citing two sources counts under both and source counts exceed the number of questions. LongDocURL's documents are born-digital.

A.5 Evaluation

Rubrics. Gemini-2.5-Flash judges every response under one of two rubrics, the first returning *correct* or *incorrect* only, while the second allows for an additional *abstained* outcome. The *accuracy rubric* (Figure 9) states only the correctness rule. The *taxonomy rubric* (Figure 10) adds a definition of abstention and assigns each incorrect response the first category of Table 8 that applies. Because it recognises refusals however they are worded, it reports more refusals, and its answerable accuracy is two to three points lower. Levels are therefore compared only within a rubric. The taxonomy rubric

Summary		By domain	
Property	**Value**	**Domain**	**Q**
Questions	1091	Research report	293
Answerable	847	Academic paper	204
1 page	480	Guidebook	156
2 pages	246	Tutorial/Workshop	139
3+ pages	112	Financial report	117
Unanswerable	244	Brochure	101
Mean pages/doc	48.3	Admin/Industry	81
Pages min–max	9–468	Total	1091

Table 6: MMLongBench-Doc composition.

Property	Value	Evidence source	Q
Questions	2,325	Text	994
Documents	396	Table	871
Mean pages/doc	85.6	Layout	779
Median pages/doc	78	Figure	548
Pages min–max	51–149	Others	8

Table 7: LongDocURL composition.

scores Tables 1, **??** and 19 and Figures **??**–13; every other result uses the accuracy rubric.

Metrics. On unanswerable questions an abstention is correct. On answerable questions it is incorrect and counts as a false refusal. Refusal rates are computed separately for the two pools.

Judge Validation. Table 9 reports agreement with two references. On 500 stratified responses, Gemini-2.5-Flash agrees with GPT-4o on 96.0% of verdicts (Cohen's $\kappa = 0.918$), and on a 120-response subset it agrees with a human annotator on 98.3% ($\kappa = 0.966$).

```
You judge answers to document questions.
Return only JSON with keys:
- verdict: one of correct, incorrect, abstained
- extracted_answer: the answer extracted from the model response, or empty string
- rationale: a short reason

Mark correct when the model answer is semantically equivalent to the gold answer.
For unanswerable questions, mark correct only when the model abstains.
```

Figure 9: Accuracy rubric.

```
⟨accuracy rubric, with an error_category key in place of rationale⟩

ABSTENTION
Return the verdict "abstained" whenever the response declines to answer because it judges the evidence insufficient, however it is worded. "Not answerable", "the document does not provide this", "there is no document evidence to determine this" and "it is not possible to determine this from the given information" are all abstentions. A response that notes a limitation and then commits to an answer anyway is NOT an abstention: judge the answer it commits to.

ERROR CATEGORY
Decide the verdict FIRST, using only the two rules above, and do not revisit it. The error category explains a verdict you have already reached; it never changes one. In particular, a response that states the gold answer in its own words, or surrounds it with explanation, is still correct.

⟨the eleven categories of Table 8, in priority order⟩
```

Figure 10: Taxonomy rubric.

Category	Assigned when
`insufficient_evidence`	The response declines: the evidence does not support an answer.
`no_answer_reached`	It reasons but never commits, trailing off or stopping mid-calculation.
`answered_unanswerable`	The question is unanswerable and a substantive answer is given.
`equivocal`	The gold answer is present but not committed to.
`wrong_count`	A "how many" question is answered with a different count.
`wrong_value`	A non-count numeric answer differs from the gold value.
`wrong_entity`	A non-numeric answer names a different thing.
`incomplete_list`	The gold answer is a set and the response gives a subset.
`over_inclusive_list`	The response includes items the gold answer does not.
`unit_or_format_mismatch`	The same quantity in a non-matching unit, scale or granularity.
`other`	None of the above.

Table 8: Error categories of the taxonomy rubric, in priority order.

Reference	n	Agree.	Cohen's κ [95% CI]
GPT-4o	500	96.0%	0.918 [0.881, 0.951]
Human annotator	120	98.3%	0.966 [0.915, 1.000]

Table 9: Judge agreement with Gemini-2.5-Flash.

	Born-digital		Scanned	
Text source	**Text**	**+Image**	**Text**	**+Image**
PaddleOCR-VL				
Text	50.9	54.9	35.4	55.4
Layout	43.8	47.5	26.3	57.9
Table	47.6	46.5	59.4	71.9
Chart	23.8	40.8	4.2	54.2
Figure	17.4	37.0	31.1	57.5
All	40.1	49.7	32.8	61.2
MinerU 2.5				
Text	30.1	49.1	30.8	53.8
Layout	20.0	43.8	18.4	50.0
Table	36.8	42.7	56.2	65.6
Chart	17.7	38.5	14.6	47.9
Figure	12.5	38.6	25.5	50.9
All	26.6	46.0	27.4	55.7
Unlimited-OCR				
Text	50.4	53.1	36.9	56.9
Layout	43.8	52.5	36.8	60.5
Table	49.7	48.1	59.4	71.9
Chart	23.8	42.3	2.1	58.3
Figure	20.7	38.0	34.0	58.5
All	42.3	50.2	34.3	63.7
PyMuPDF				
Text	50.0	54.4	3.1	50.8
Layout	40.0	47.5	2.6	52.6
Table	48.1	44.9	0.0	59.4
Chart	24.6	37.7	0.0	54.2
Figure	19.6	41.3	7.5	52.8
All	40.6	49.7	4.0	57.7

Table 10: Accuracy by text source, evidence source and scan status. Accuracy (%) on answerable MMLongBench-Doc questions with gold pages, with the text alone and with the page image added.

	TLV			V		
Source	`low`	`med`	`high`	`low`	`med`	`high`
Text	52.9	55.0	55.0	38.8	45.4	51.5
Layout	46.6	50.0	54.2	47.5	49.2	53.4
Table	48.8	49.8	48.8	25.3	37.3	42.4
Chart	38.2	44.9	48.3	34.3	44.9	48.3
Figure	39.7	45.5	48.3	38.3	40.0	43.8
All	49.2	52.5	54.7	37.8	45.7	50.2

Table 11: Accuracy by image resolution and evidence source. Accuracy (%) on answerable MMLongBench-Doc questions with gold pages, with parser text (TLV) and without it (V).

Task and question type	**T**	**TL**	**TLV**	**V**	n
Understanding (`extract`)	74.1	77.1	**83.4**	74.6	1,243
Locating	43.5	**51.6**	50.0	45.4	695
`extract_fig2tab`	58.4	**71.6**	62.3	55.4	231
`topic2title`	27.9	28.8	**35.1**	30.8	201
`summary2title`	67.2	**81.1**	77.2	74.3	137
`summary2tab`	14.9	18.3	**19.6**	19.0	126
Reasoning	66.8	67.5	**73.4**	58.3	387
`calculate`	61.0	66.7	**72.8**	50.7	145
`count`	57.9	56.2	**66.3**	58.0	117
`compare`	**81.1**	78.8	80.2	65.2	112
`summarize`	**84.6**	76.9	81.8	**84.6**	13
All	63.7	68.0	**71.9**	63.1	2,325

Table 12: LongDocURL accuracy by task and question type. Accuracy (%) on answerable questions with gold pages; n is the number of questions.

B Additional Results

This appendix extends the results of §5–§**??**. Unless stated otherwise, results use Qwen3-VL-8B at **TLV** with gold pages on MMLongBench-Doc.

B.1 Representation

Parsers by Evidence Source. Table 10 breaks the parser comparison of Figure 2 down by evidence source. The page image helps most where text extraction loses the evidence: chart and figure questions gain 13 to 26 points on born-digital documents under every text source, while table questions change by under 6 points and lose slightly under three of the four. On scanned documents embedded text answers almost nothing, and the image recovers 45 points or more for every source.

Resolution by Evidence Source. Table 11 extends the resolution sweep to all evidence sources. From `low` to `high`, **V** gains 12.4 points overall and 17.1 on tables, against 5.5 and 0.0 for **TLV**. Chart questions are the exception: they gain about 10 points under **TLV** as well, and parser text adds nothing to them at `med` or `high`.

LongDocURL by Question Type. Table 12 splits the LongDocURL results of Table **??** by question type. Accuracy varies far more across question types than across representations: `summary2tab` stays below 20% under every representation, while `extract` never falls below 74%. Parser text alone is the best representation for two of the locating types, and **V** is weakest on `calculate` and `compare`, which depend on exact values.

B.2 Selection

Retrieval Recall. Figure 11 compares three text and three visual retrievers. Visual retrievers lead at every depth, and ColQwen3 is the strongest, which is why it orders pages in the selection experiments. Text retrieval over the embedded text layer collapses on scanned documents, and ranking parser text instead recovers much of the gap. Even ColQwen3 leaves gold pages outside the top ten,

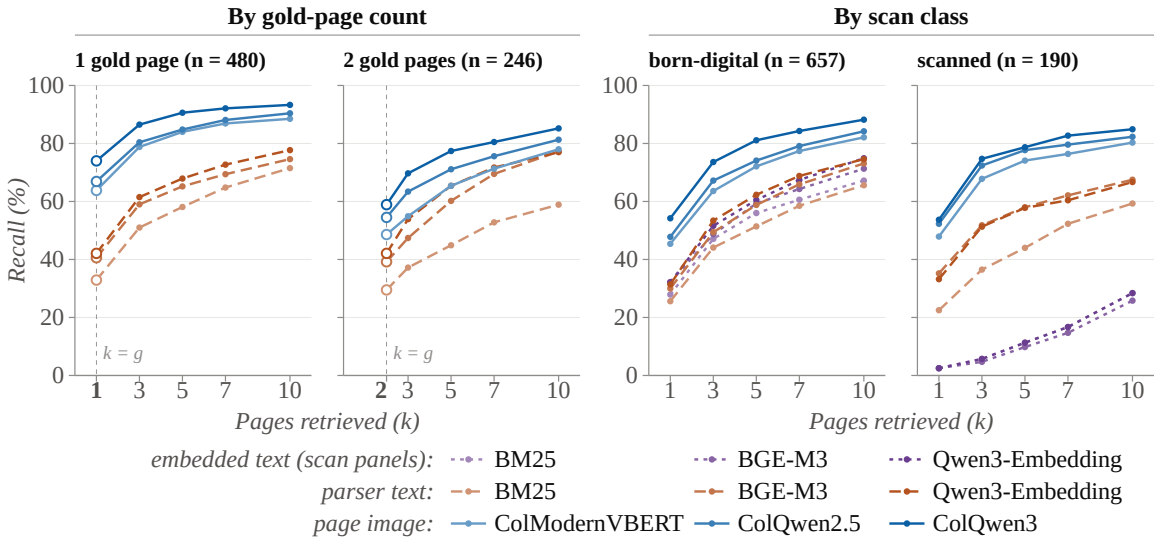


Figure 11: Page-level recall on MMLongBench-Doc by retrieval depth, split by gold-page count and by scan status.

	Drop top		Drop bottom		Keep top		Keep bottom		
Source	−	+	−	+	−	+	−	+	n
Text	48.1	2.1	15.8	4.3	19.3	3.7	52.3	2.4	621
Layout	36.5	1.6	20.1	2.7	23.4	2.3	40.7	1.8	513
Table	32.2	2.6	15.9	3.9	17.2	4.3	33.0	1.7	233
Figure	44.2	2.8	22.7	4.4	26.3	5.6	48.2	2.4	251
All	43.2	2.0	19.6	3.6	22.2	3.5	46.5	1.8	1,093

Table 13: LongDocURL verdict transitions after removing or retaining gold pages. Share (%) of multi-page questions whose TLV verdict moves from correct to incorrect (−) or incorrect to correct (+) against the complete gold set; n is the paired question count.

more often for questions with two gold pages.

Coverage on LongDocURL. Table 13 repeats the coverage experiment of Figure 3 on LongDocURL. Omission again breaks far more answers than it repairs. Unlike on MMLongBench-Doc, retrieval rank is informative here: removing the highest-ranked gold page breaks 43.2% of answers, against 19.6% for the lowest-ranked page.

Distractor Transitions by Step. Table 14 gives the paired transitions behind Figure 4, comparing each depth with the one before it. The first three added pages break 12 to 14% of answers for both reasoners and both question groups, well above any later step (at most 7.9%). Every step also repairs answers, at 2 to 9%, so the accuracy curve is the net of two opposing movements. After the first step the two roughly cancel for Qwen3-VL-8B, while broken answers continue to outnumber repaired ones at most steps for Gemma-3-12B on single-hop questions.

		+3	+5	+7	+10	+13	+15	+18
Qwen3-VL-8B								
Single-hop	broken	12.4	5.4	5.4	4.8	6.6	4.1	3.8
	repaired	8.7	5.4	3.1	4.2	5.7	3.2	2.3
Multi-hop	broken	14.4	4.9	6.8	7.9	5.2	5.0	5.9
	repaired	5.6	6.5	7.7	4.3	4.1	5.9	3.2
Gemma-3-12B								
Single-hop	broken	13.7	7.1	6.5	7.9	6.0	7.1	5.2
	repaired	6.0	3.7	4.4	4.5	6.7	5.7	2.8
Multi-hop	broken	14.0	7.0	4.5	5.3	7.1	5.0	6.8
	repaired	5.7	6.0	5.3	5.3	3.8	3.6	5.5

Table 14: Verdict transitions at each step of the distractor sweep. Share (%) of questions whose TLV verdict moves from correct to incorrect (broken) or incorrect to correct (repaired) between one depth and the next, on MMLongBench-Doc; – marks a step that has not completed.

Distractors on LongDocURL. Table 15 repeats the distractor sweep on LongDocURL with Qwen3-VL-8B. This experiment is still running, and the table reports the depths completed so far. On single-hop questions the first three added pages lower accuracy from 77.2% to 68.5%, breaking 13.5% of answers and repairing 4.8%, close to the first step on MMLongBench-Doc.

B.3 Reasoning

Further Reasoner Configurations. Table 16 adds two comparisons to Table **??**. Gemma-3 improves with scale at every representation, as Qwen3-VL does, reaching 52.0% at **TLV** with 27B. The Thinking variant of Qwen3-VL-8B reaches 62.2% at **TLV**, 9.8 points above the Instruct model at the same weight footprint and above the 32B Instruct model.

		+0	+3	+5	+7	+10	+13	+15	+18
Single-hop	accuracy	77.2	68.5	67.3	69.5	65.6	68.1	66.8	67.3
	broken	–	13.5	6.2	4.0	6.2	3.1	5.1	5.6
	repaired	–	4.8	5.1	5.9	2.5	5.6	3.7	6.2
Multi-hop	accuracy	66.2	61.4	58.8	56.3	56.2†	55.0†	57.6†	56.6†
	broken	–	11.2	8.1	7.3	6.2†	4.9†	4.3†	6.0†
	repaired	–	6.5	5.6	5.0	5.9†	5.4†	6.8†	5.1†

Table 15: LongDocURL accuracy and verdict transitions under added non-gold pages. Accuracy (%) at TLV given every gold page plus the k highest-ranked non-gold pages, with the share of questions broken or repaired against the previous depth; – marks a depth with nothing judged yet. † marks a depth still running, read on the questions judged so far, which are fewer and easier than a finished depth's.

Reasoner	T	TL	TLV	V	GB
Scale (Qwen3-VL, bf16)					
2B	24.7	31.9	38.3	31.9	4.3
4B	32.3	38.6	50.4	42.4	8.9
8B	31.9	38.4	52.4	45.5	17.5
32B	35.5	43.9	59.6	56.7	66.7
Reasoning variant (Qwen3-VL-8B)					
Instruct	31.9	38.4	52.4	45.5	17.5
Thinking	34.8	42.9	62.2	52.1	17.5

Table 16: Accuracy for further reasoner configurations. Accuracy (%) on answerable MMLongBench-Doc questions with gold pages; GB is the weight footprint.

Gemma-3 Scale by Hop. Table 17 repeats the scale comparison of Figure 5 for Gemma-3. Multi-hop accuracy rises from 31.9% at 4B to 43.3% at 27B, and single-hop accuracy rises alongside it, so the gap between them does not narrow (19.6 points at 4B, 22.6 at 27B), as with Qwen3-VL.

Prompt Modes by Hop. Table 18 splits the prompt modes by evidence structure. The near-zero overall effect of `cot` reported in §5.3 hides two opposite ones: it raises multi-hop accuracy by 3.1 points and lowers single-hop accuracy by 2.8. `abstain` costs accuracy on both, and multi-hop questions are refused more often (28.1% against 22.0%). `abstain_cot` recovers most of that loss on both.

Prompt Modes by Evidence Source. Table 19 reports six prompt modes by evidence source under the taxonomy rubric. Its refusal rates are higher than those of Table 18 and Figure 6, which use the accuracy rubric, but the pattern is the same: `abstain` raises unanswerable refusal from 45.5% to 74.6% and answerable refusal from 6.2% to 27.7%, with figure and table questions refused most. Adding a sentence against declining supported questions (`abstain_balanced`) recovers little, and `grounded` alone lowers accuracy without improving refusal.

Reasoner	Single-hop	Multi-hop	Δ
Gemma-3-4B	51.5	31.9	19.6
Gemma-3-12B	61.7	39.0	22.7
Gemma-3-27B	65.9	43.3	22.6

Table 17: Gemma-3 accuracy on single- and multi-hop questions by model size. Accuracy (%) at TLV on answerable MMLongBench-Doc questions with gold pages; Δ is the multi-hop deficit.

	Single-hop		Multi-hop	
Prompt mode	Acc.	Ref.	Acc.	Ref.
`none`	**66.4**	5.7	44.9	5.4
`grounded`	59.2	6.0	38.5	4.1
`abstain`	53.7	22.1	34.7	28.5
`abstain_balanced`	55.1	17.5	35.7	24.2
`cot`	63.6	5.4	**48.0**	7.2
`extract_cot`	59.2	9.2	44.6	8.2
`abstain_cot`	63.1	16.3	43.9	22.1

Table 18: Prompt-mode results on single- and multi-hop questions. Accuracy and refusal (%) at TLV on answerable MMLongBench-Doc questions with gold pages.

B.4 Error Analysis

The taxonomy rubric assigns each incorrect response one error category (Table 8). Figures 12 and 13 show the share of each category, with responses that commit to no answer on the left and wrong answers on the right.

By Prompt Mode. The abstention instructions change the kind of error more than its amount (Figure 12). Under `abstain`, refusals on answerable questions rise from 6.1% to 27.6% while every wrong-answer category shrinks, and answers to unanswerable questions fall from 53.7% to 25.4%. `grounded` moves in the opposite direction, removing unfinished responses but adding wrong values and counts.

By Evidence Source. Wrong counts dominate layout and figure questions, at 28.8% and 24.1% (Figure 13). Table and chart questions fail mainly on values, and table questions most often end without stating an answer.

	Answerable accuracy						Answerable refusal						Unans.
Prompt mode	**Text**	**Layout**	**Table**	**Chart**	**Figure**	**All**	**Text**	**Layout**	**Table**	**Chart**	**Figure**	**All**	**refusal**
none	**55.3**	**50.8**	48.4	**45.5**	44.8	52.5	7.3	0.8	6.7	7.3	7.6	6.2	45.5
grounded	51.2	42.4	40.1	33.7	42.4	45.7	7.0	0.0	7.2	5.1	7.2	6.0	37.3
abstain	45.7	41.5	39.2	28.7	34.5	41.4	**24.1**	**19.5**	**30.3**	**25.3**	**35.5**	**27.7**	**74.6**
abstain_balanced	47.4	41.5	37.3	30.9	36.2	42.6	19.2	18.6	25.5	18.5	31.4	23.2	68.9
cot	54.6	50.0	**55.3**	**45.5**	**45.2**	**53.1**	7.7	3.4	7.2	9.6	6.9	6.6	43.0
extract_cot	51.5	**50.8**	52.5	39.3	44.1	50.2	9.9	2.6	9.6	6.3	11.1	8.8	47.7

Table 19: Prompt-mode results at TLV. Rates (%) on MMLongBench-Doc: accuracy and refusal on answerable questions with gold pages, by evidence source, and refusal on unanswerable questions with BM25's top three pages, where refusing is correct.

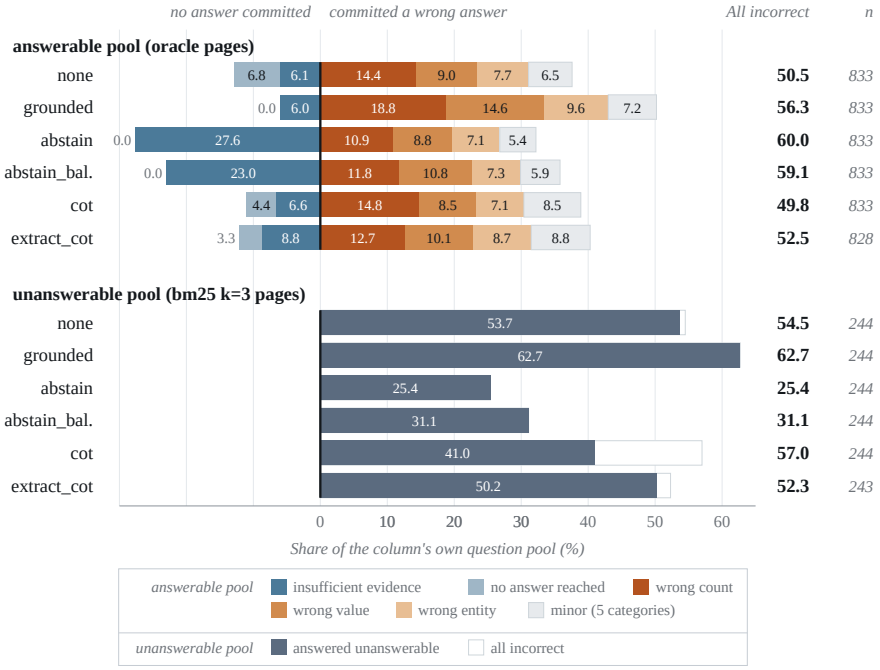


Figure 12: Error categories by prompt mode on answerable and unanswerable questions.

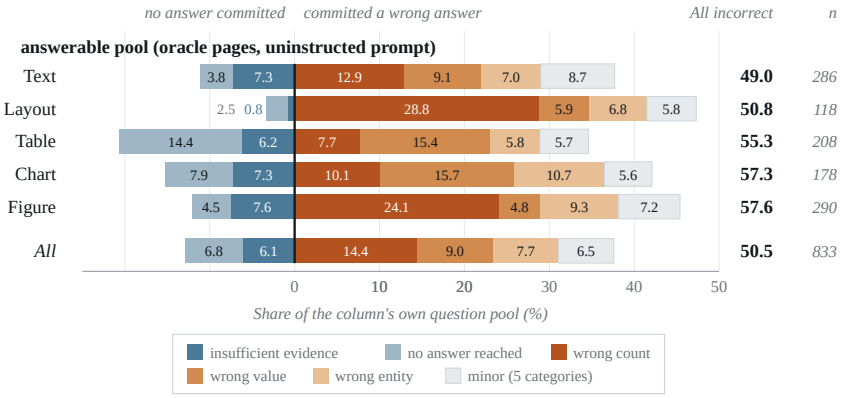


Figure 13: Error categories by evidence source on answerable questions with no instruction.

C Worked Examples

This appendix shows individual questions behind the rates of §5, at least one for each failure mechanism of §3. Each box gives the question, the gold answer, and the reasoner's output under every condition shown, marked ✓ correct, × incorrect or ○ abstained. The reasoner is Qwen3-VL-8B at **TLV** with no instruction unless a row says otherwise, and verdicts follow the accuracy rubric. Page numbers are zero-based indices into the document. None of these questions is one of the six in Figure 7.

C.1 Representation

R1 Modality ceiling `mmlongbench:000014`

Presentation deck (Tutorial/Workshop), born-digital. Gold page 13, a single chart. Evidence source: chart.

Q According to the chart on page 14 how much time was spent with family and friends in 2010?
Gold 21%

The page is one chart, and the parser has nothing to transcribe. PaddleOCR-VL emits only a placeholder:

```
<div style="text-align: center;"><img src="[. . .]"
alt="Image" width="52%" /></div>
```

T	○	"The document does not provide any evidence or chart on page 14, so it is not possible to determine how much time was spent with family and friends in 2010 [. . .]"
TL	○	"[. . .] The image referenced (on page 14) is not described or transcribed, so its content cannot be interpreted. [. . .]"
V	✓	"According to the chart on page 14, in 2010, 21% of weekend time was spent with family and friends."
TLV	✓	"According to the chart on page 14, in 2010, 21% of weekend time was spent with family and friends."

The gold page reaches every representation, so nothing is missing from the context. The value lives in the chart's geometry and no text channel carries it, so both text representations decline instead of guessing, which is the refusal pattern of Table 1.

R2 Conversion fidelity: parser `mmlongbench:000513`

Guidebook, born-digital. Gold page 9, a numbered list of market names with a code beside each. Evidence source: text and table.

Q How many geographic market names are listed under Los Angeles area?
Gold 13

The list runs from item 14 to item 26. The three parsers keep its text and differ in what they do with the code column:

```
PaddleOCR-VL: California 14-26 | 14. San Fernando Valley
(West) CA14 | [. . . ] | 26. Santa Ana CA26
MinerU 2.5: 14. San Fernando Valley (West) | [. . . ] |
26. Santa AnaEPS | Code | 4.14 | [. . . ] | 4.26
Unlimited-OCR: <table><tr><td>14.</td><td>San Fernando
Valley (West)</td><td>CA14</td></tr> [. . . ]
```

T	✓	"The document lists **13** geographic market names under the Los Angeles area. [. . .]"
TL, PaddleOCR-VL	×	"26 The document lists 26 geographic market names under the 'Los Angeles Area' section, ranging from CA14 to CA26."
TL, MinerU 2.5	×	"26 The document lists 26 geographic market names [. . .] ranging from 'San Fernando Valley (West)' (item 14) to 'Santa Ana' (item 26)."
TL, Unlimited-OCR	✓	"There are **13** geographic market names listed under the Los Angeles area. The list includes entries numbered 14 through 26 [. . .]"
TLV, PaddleOCR-VL	✓	"[. . .] there are **13** geographic market names listed under the Los Angeles area. [. . .] codes from 14 to 26. Counting these entries [. . .]"

Every parser returns all thirteen names, yet two flatten the list so that its last label reads as its length. The parser that keeps the rows as a table gives the right count, as does the same flattened parse once the page image accompanies it: the image is the redundant channel that absorbs the conversion error.

R3 Conversion fidelity: resolution `mmlongbench:000065`

Academic paper, born-digital. Gold page 6, a results table with one row per model and temperature. Evidence source: text and table.

Q From the paper, which temperature gives ChatGPT the highest alignment score?
Gold 0.1

V, `low`	×	"[. . .] the temperature that gives ChatGPT the highest alignment score is **0.5**."
V, `med`	×	"[. . .] is **0.5**. [. . .] Table 3 lists three ChatGPT models: ChatGPT (0.5), ChatGPT (0.1), and ChatGPT (0.9) [. . .]"
V, `high`	✓	"[. . .] the temperature that gives ChatGPT the highest alignment score is **0.1**."
TLV, `low`	✓	"[. . .] ChatGPT (0.1): 85.9, ChatGPT (0.5): 84.5, ChatGPT (0.9): 84.1. [. . .] **85.9** is the highest alignment score, which corresponds to [. . .] **0.1**."

The image alone finds the right table at every resolution but only reads its small digits correctly at `high`*. With parser text the values are available as characters, so the lowest resolution already suffices, as in the table rows of Table 11.*

C.2 Selection

S1 Evidence coverage `mmlongbench:000082`

Survey paper (Academic paper), born-digital. Gold pages 8 and 9: the sentence that answers the question starts on the last line of page 8 and finishes on page 9. `drop_top` leaves page 8 and `drop_bottom` page 9. Evidence source: text.

Q Which method integrates knowledge into white-box models via directive fine-tuning?
Gold PKG

gold pages	✓	"[. . .] the method that integrates knowledge into white-box models via directive fine-tuning is **PKG**. [. . .] 'PKG introduces an innovative method for integrating knowledge into white-box models via directive fine-tuning [75].' [. . .]"
`drop_top`	×	"[. . .] the method that integrates knowledge into white-box models via directive fine-tuning is **Fine-tuning Embedding Model**. [. . .]"
`drop_bottom`	×	"The method that integrates knowledge into white-box models via directive fine-tuning is **directive fine-tuning**. The document explicitly states: 'introduces an innovative method for integrating knowledge into white-box models via directive fine-tuning [75].' [. . .]"

Page 8 ends on the method's name and page 9 opens with what it does, so neither page holds the answer. With one of them withheld the reasoner does not decline: it names a nearby section heading, or repeats the question's own phrase. Both removals break the answer, so retrieval rank does not identify a page that can safely be omitted.

S2 Distractor exposure: broken `mmlongbench:000078`

Survey paper (Academic paper), born-digital, multi-hop. Gold pages 5 (a table of methods) and 16 (the reference list). +3 adds pages 0, 6 and 7; page 0 is the survey's own title page. 4-bit weights.

Q What is the paper's full title that proposes the method that has a retrieval granularity of phrase?
Gold Copy is all you need.

gold only	✓	"[. . .] the method with a retrieval granularity of "Phrase" is **CoG [29]**. [. . .] the full title of the paper proposing the CoG method is: **"Copy is all you need"**. This is found in the References section [. . .]"
+3	×	"[. . .] the paper's full title is **Retrieval-Augmented Generation for Large Language Models: A Survey**. This is explicitly stated in the title section of the document [. . .]"
+5 to +18	×	The same answer at every completed depth.

Both gold pages are still present and nothing in them changed. The added title page is a plausible partial match for "the paper's full title", and the reasoner takes it instead of following the table entry to its reference. Coverage is complete; the failure is in using the evidence amid competing context.

S3 Distractor exposure: repaired `mmlongbench:000243`

Rezoning study (Research report), born-digital, single-hop. Gold page 7, which carries a table of residential capacity by borough. +3 adds pages 4, 8 and 12, the surrounding discussion. 4-bit weights.

Q What is the residential capacity of Staten Island from 2003 to 2007? Give me an integer.
Gold 435000000

gold only	×	"80000"
+3	✓	"435000000"
+10	✓	"435000000"
+13	×	"5980000"
+18	✓	"435000000"

The table holding the answer is supplied in every row. The reasoner misreads it in isolation and reads it correctly once neighbouring pages of the same section are present, then fails again at one deeper setting. Added pages therefore both repair and break answers on the same question, the two-way movement behind the transitions of Figure 4.

Distractors on LongDocURL. Table 15 repeats the distractor sweep on LongDocURL with Qwen3-VL-8B. This experiment is still running, and the table reports the depths completed so far. On single-hop questions the first three added pages lower accuracy from 77.2% to 68.5%, breaking 13.5% of answers and repairing 4.8%, close to the first step on MMLongBench-Doc.

C.3 Reasoning

N1 Evidence integration `mmlongbench:001015`

Annual report (Financial report), born-digital, multi-hop. Gold pages 35 (the income statement) and 39 (the cash flow statement). Rows are Qwen3-VL sizes. Evidence source: table.

Q what is EBITDA for costco in FY2021?
Gold 8489

2B	×	"[. . .] EBITDA for Costco in FY2021 is $6,708 million. This is found in the [. . .] row labeled 'Operating income' [. . .]"
4B	×	"The EBITDA for Costco in FY2021 is $6,708 million. [. . .] EBITDA [. . .] is typically calculated as Operating Income plus Depreciation and Amortization, but in this document, the 'Operating income' row is presented as the direct measure of EBITDA [. . .]"
8B	×	"[. . .] Operating Income + Depreciation and Amortization + Interest Expense + Provision for Income Taxes = $6,708 + $1,781 + $(171) + $1,601 = **$8,469 million**"
32B	✓	"[. . .] From the Consolidated Statements of Income: Operating Income = $6,708 million. From the Consolidated Statements of Cash Flows: Depreciation and Amortization = $1,781 million. [. . .] $6,708 + $1,781 = **$8,489 million**"

Every model receives the same two statements. The two smallest stop at operating income on the first page, one of them after stating the formula that needs the second. The 8B reaches the cash flow statement but adds lines that operating income already excludes, and only the largest joins the two operands. The evidence was sufficient throughout; what scales is the ability to combine it.

N2 Response calibration: answerable
`mmlongbench:000418`

E-commerce market deck (Research report), scanned, multi-hop. Gold pages 11 and 14, one bar chart on each. Evidence source: chart.

Q How many people in India were using a debit card when the e-commerce sales reached 11 billion?
Gold 399 million

`none`	✓	"[. . .] when e-commerce sales reached $11 billion (in 2014), the number of debit card users in India was **399 million**. [. . .]"
`cot`	✓	"Step 1: [. . .] the $11 billion mark is shown for the year 2014. Step 2: [. . .] the bar chart titled 'Number of Debit Card users in India (In millions)' shows the value for 2014 as 399 million. [. . .] Answer: 399 million"
`abstain`	○	"Not answerable"
`abstain_cot`	✓	"Step 1: [. . .] e-commerce sales were $11 billion in 2014. Step 2: [. . .] For 2014, the number is 399 million. [. . .] Answer: 399 million"

The answer is on neither page alone: one chart gives the year, the other the count for that year. Under `abstain` *the reasoner refuses a question it answers correctly with no instruction, a false refusal. Asking it to reason before deciding makes it locate both values first, and the refusal disappears.*

N3 Response calibration: unanswerable
`mmlongbench:000047`

Data-leadership talk (Tutorial/Workshop), scanned, unanswerable. BM25's top three pages (0, 1, 2) are supplied; the document mentions no video.

Q Which Youtube does the slides use to show the consequce of having a weak data leadership?
Gold Not answerable

`none`	×	"The slides use a YouTube clip from the TV show *Breaking Bad* to illustrate the consequence of having weak data leadership [. . .] as symbolized by the character Walter White hoarding money [. . .]"
`cot`	✓	"[. . .] Step 4: However, the document evidence does not mention any YouTube or reference any video content. The image is a still image, not a YouTube video. [. . .] Answer: Not specified in the document."
`abstain`	✓	"Not answerable"
`abstain_cot`	✓	"[. . .] Step 3: The document does not mention YouTube at all. [. . .] Answer: Not answerable"

One supplied page shows a man holding stacks of money, and with no instruction the reasoner builds a show, a character and an interpretation out of it. Each of the other prompts leads it to check the premise against the pages and decline, which is the correct response here.